

Cuauhtemoc Aguilar Mandujano

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with experience building production-ready applications from schema design to user interface, using TypeScript, Java, and Python. Strong foundation in object-oriented design, data structures, REST API development, and agile collaboration.

SKILLS

Programming: TypeScript, JavaScript, Java, Python (Pandas, NumPy)

Web & Databases: Node.js, REST APIs, PostgreSQL, MySQL, SQLite, Drizzle ORM

Concepts: Object-Oriented Design, System Design, Data Structures & Algorithms, Data Modeling

Data & Analysis: SQL, Data Visualization, Tableau, Power BI, Statistical Analysis

Tools: Git, GitHub, Excel

PROJECTS

Auto Guesser <https://auto-guesser.vercel.app/>

September 2026

Next.js, TypeScript, PostgreSQL, Naive Bayes, REST APIs

- Engineered a full-stack web app with Next.js and TypeScript, using serverless API routes to handle game logic and crowdsourced data ingestion.
- Built a custom Naive Bayes decision engine that recalculates entropy and probabilities every turn, minimizing the number of questions needed to correctly identify an unknown vehicle.
- Designed a normalized PostgreSQL database (Neon, Drizzle ORM) supporting 100+ vehicles and 60+ attributes, built to accommodate ongoing crowdsourced expansion without schema changes.
- Integrated the Wikimedia API with a database-level caching layer to serve entity imagery, eliminating redundant network requests and cutting load times by approximately 15%.

Dimension Shift https://michi9450.github.io/491_Time_Platformer/

March 2026

JavaScript, HTML, CSS, Git, GitHub

- Engineered core gameplay systems (approximately 2,000+ LOC) including player physics, collision detection, camera smoothing, and event-driven input handling in a custom JavaScript platformer engine.
- Implemented modular game entity mechanics and an audio management system supporting sound effects and player mute controls across 10 playable levels.
- Collaborated with a 4-person team using GitHub and week-long sprint cycles, integrating features and debugging gameplay systems in an agile workflow.

EXPERIENCE

Freelance AI Trainer

May 2026 – Present

DataAnnotation

- Evaluated and optimized AI-generated Python code through rigorous code review, ensuring syntax accuracy, logical soundness, and adherence to industry standards.
- Completed 300+ tasks creating comprehensive test cases and debugging algorithms to train models in code generation, edge-case handling, and runtime efficiency.
- Benchmarked model outputs against strict technical guidelines, grading responses for factual accuracy, security vulnerabilities, and structural integrity.

Fellow

August 2024 – April 2025

Code2040 Fellows Program

- Selected for a nationally competitive program supporting early-career technologists from underrepresented backgrounds in software engineering.
- Participated in technical workshops and collaborative projects focused on software engineering and industry practices.

EDUCATION

University of Washington Tacoma

Tacoma, WA

Bachelor of Arts in Computer Science and Systems. Minor in Business Data Analytics.

March 2026

Relevant Coursework: Software Development and Quality Assurance, Data Structures and Algorithms, Database System Design, Predictive Analytics, Business Intelligence, Machine Learning, Principles of Web Design, User Centered Design